Yiran Lei

Ph.D. Student in Computer Science, Carnegie Mellon University

yiranlei@andrew.cmu.edu yiranlei.com GitHub

Education

- **Carnegie Mellon University** Ph.D. in Computer Science — Advisor Justine Sherry — 2023–Present
- **Tsinghua University** M.E. in Computer Science — Advisor Mingwei Xu — 2020–2023
- **Tsinghua University** B.E. in Software Engineering — 2016–2020

Selected Honors

- **CMU SCS Fellowship in Computer Science** — 2026
- **China National Scholarship** — 2022
- **Fellowship for Comprehensive Excellence (Second Class), Tsinghua** — 2021

Publications

- **FAST: An Efficient Scheduler for All-to-All GPU Communication** — NSDI 2026
  Yiran Lei, Dongjoo Lee, Liangyu Zhao, Daniar Kurniawan, Chanmyeong Kim, Heetaek Jeong, Changsu Kim, Hyeonseong Choi, Liangcheng Yu, Arvind Krishnamurthy, Justine Sherry, and Eriko Nurvitadhi.
- **SwitchNIC: An Hybrid Architecture for Network Functions with Fast and Consistent Shared State** — CoNEXT 2025
  Yiran Lei, Francisco Pereira, Arvind Krishnamurthy, and Justine Sherry.
- **PrintQueue: Performance Diagnosis via Queue Measurement in the Data Plane** — SIGCOMM 2022
  Yiran Lei, Liangcheng Yu, Vincent Liu, and Mingwei Xu.
- **DOVE: Diagnosis-driven SLO Violation Detection** — ICNP 2021
  Yiran Lei, Yu Zhou, Yunsenxiao Lin, Mingwei Xu, and Yangyang Wang.

Preprint

- **The Energy Cost of Execution-Idle in GPU Clusters** — arXiv 2026
  Yiran Lei, Jared Fernandez, Vasilis Kypriotis, Dimitrios Skarlatos, Emma Strubell, Justine Sherry, and Daniel Vosler.

Patents

- **Methods and Systems for All-to-All Communications in Distributed Computing Systems** — WO2026161384A1
  Yiran Lei, Daniar Kurniawan, Heetaek Jeong, and Eriko Nurvitadhi. — MangoBoost Inc. · July 30, 2026

Research Experience

- **Microsoft Research** Research Intern — Mentor Michael Papamichael — May–Aug 2025
- **MangoBoost** Technical Intern — Mentor Eriko Nurvitadhi — May 2024–Feb 2025
- **University of Washington** Visiting Research Intern — Advisor Arvind Krishnamurthy — Sep 2022–May 2023
- **University of Pennsylvania** Research Intern — Advisor Vincent Liu — Sep 2021–Aug 2022
- **UCLA** Visiting Research Intern — Advisor Lixia Zhang — Jun–Aug 2019

Invited Talks

- **Princeton University** — Hosted by Maria Apostolaki — Nov 2026
- **University of Pennsylvania** — Hosted by Vincent Liu — Apr 2026

Teaching

- **Carnegie Mellon University** Teaching Assistant · Computer Networks and the Internet (15-441/641) — Fall 2025, Fall 2026
- **Tsinghua University** Teaching Assistant · The Principle of Computer Network (40240513) — Fall 2021